

Optical Microscopy

Since Sorby published his observations on the structures of steels in 1863, optical microscopes have become one of the most widely used and versatile instruments for examining the structures of engineering materials. Despite its extensive use, it has also been the most misused, abused, and misunderstood scientific instrument, for it will produce an image of a sort no matter how badly it is used. To use it effectively, even in its simplest applications, a knowledge of the simple theory of the microscope is necessary. Thus my aim has been to give a simple and, where possible, quantitative account of both the theory and the use of the microscope, including the various special techniques for which it can be used. But, no matter how effectively the microscope is used, if the specimen is inadequately prepared, the results of examination will be of doubtful value. Therefore, we will include a section on specimen preparation that sets out the range of methods available, and describes the methods that are most widely used, in sufficient detail to enable the reader to carry them out.

1. The mechanical construction of the microscope

There are two types of microscope, simple and compound. A simple microscope is a magnifying glass and usually is designed to produce an image that is up to several times larger than the object. These are useful for visual examination of macroscopic features, e.g. of cracks and fractures. However, to observe and resolve finer detail a compound microscope must be used.

Compound microscope consists of two compound lenses, the objective and the eyepiece. The former produces a magnified real image of the object, that is further magnified by the latter. However, since most objects are not luminous, an illuminating system needs to be included in the apparatus and is often incorporated in the microscope unit. The type of illuminating system used depends on whether the object is to be viewed by transmitted or reflected light. Compound microscopes can be divided into two groups; bench microscopes, intended mainly for direct visual examination of the image and projection microscopes, intended for both direct visual examination and photographic recording of the image.

1.1 Bench microscope

The lenses of a compound microscope are mounted at the opposite ends of the microscope body tube (Figure 1), which is painted matt black inside to minimise internal reflections that would contribute to glare. On modern microscopes the tube is usually of fixed length, the mechanical tube length, because objectives are designed to work best at a fixed mechanical tube length. Most microscope makers have standardized on a mechanical tube length of 160 mm. Figure 1 summarizes the relationships between optical and mechanical distances in the microscope. Note that the mechanical tube length usually differs markedly from the optical tube length, and should not be substituted for the latter in optical calculations. The microscope is mounted on a heavy, well-balanced, rigid stand to minimize vibrations that would cause an unsteady image (Fig. 2).

In transmitted-light microscopes the body contains the focusing mechanism, but in reflected-light microscopes, usually called metallurgical microscopes, the mechanism more commonly is fitted to the stage. In modern bench microscopes, the limb and foot are integral, and the body, no longer a simple tube, is rigidly fixed to the stand. A prism in the body reflects the light into the eyepiece, that is at an angle of about 135° to the lower part of the body, which is vertical, eliminating the need to tilt the limb (Fig. 2a). However, in order to maintain the appropriate mechanical tube length there is included in the optical system a tube lens that may make a small contribution to the overall magnification. For metallographic work an inverted bench microscope is sometimes used (Fig. 2b). In this case the body of the microscope is arranged so that the objective lens

points upwards and a prism in the body reflects the light from the specimen into an eyepiece inclined at about 45° to the optic axis of the microscope. The stage is above the objective lens and focusing is achieved by moving the stage. The light source is built into the foot of the microscope. This design of microscope enables larger, heavier specimens to be examined than can be accommodated on the stage of a microscope of conventional design.

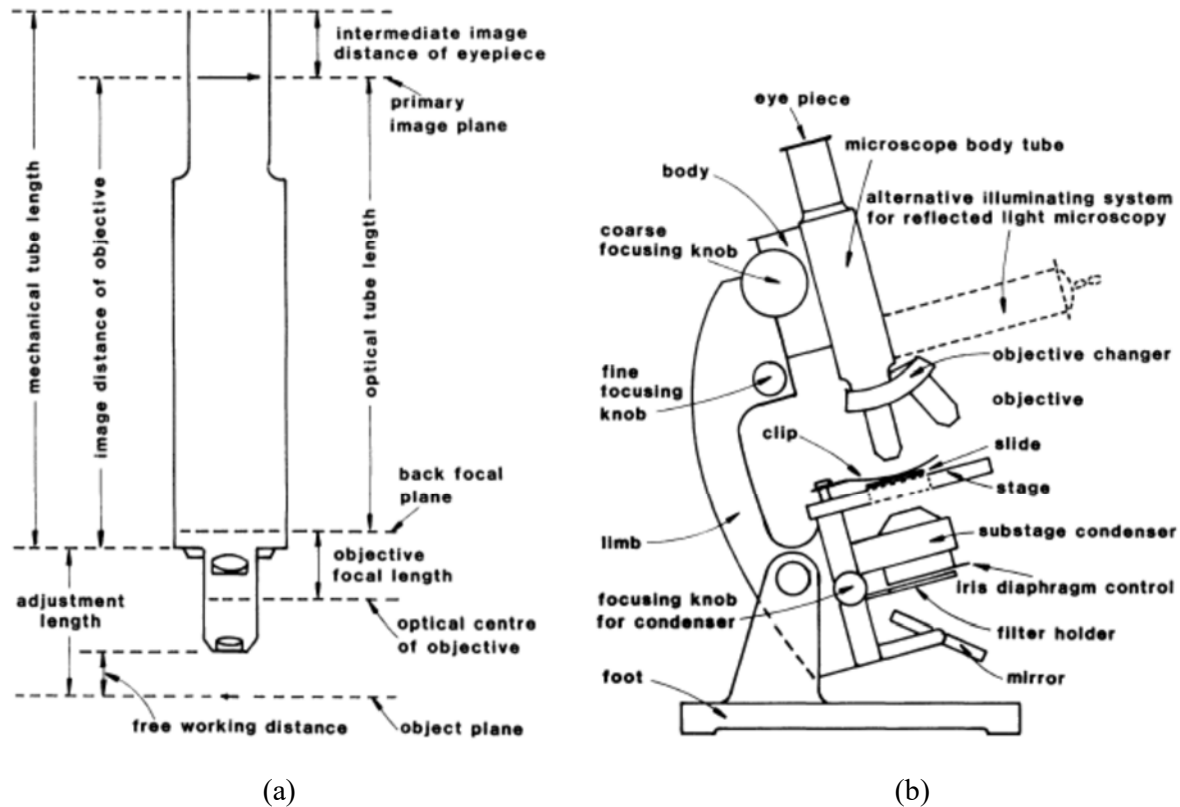


Figure 1 (a) optical and mechanical distances in the compound microscope. (b) Schematic of the components of a bench microscope.

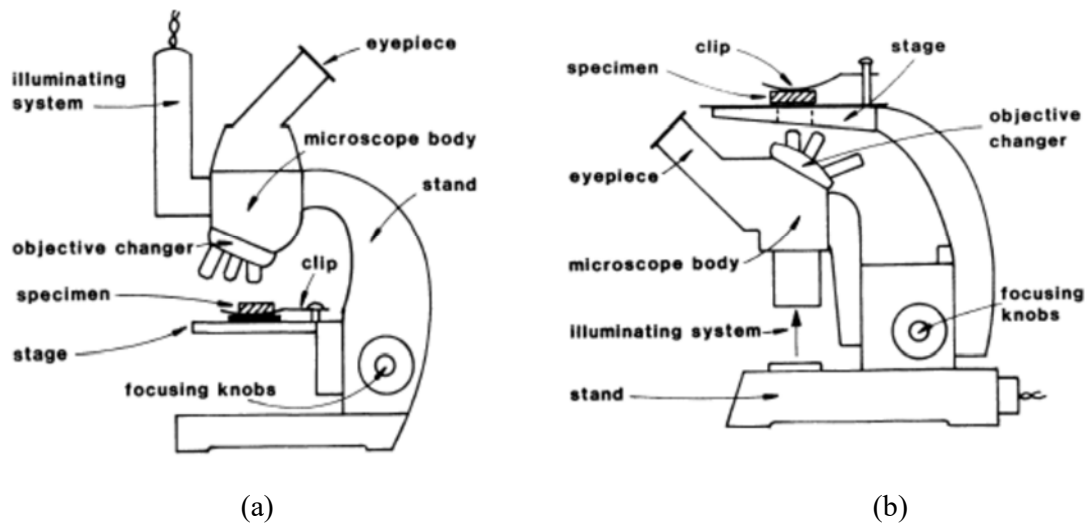


Figure 2 (a) A modern design of bench metallurgical microscope. (b) Schematic of an inverted metallurgical microscope

2. Image formation in the optical microscope

To understand the formation of the image in the microscope we must consider it from the points of view both of geometrical optics and of the wave theory of light. The former yields information about magnifications that can be attained and the types and positions of the image formed, while the latter yields information about the detail that can be resolved by the microscope objective and the quality of the image.

2.1 Geometrical optics of the compound microscope

In geometrical optics we make use only of the simple idea that light travels in straight lines unless it is refracted or reflected. Image formation in the compound microscope is most easily understood by considering the microscope to be composed of two thin convex lenses, one (OO) acting as the objective and the other (EE) as the eyepiece (Figure 3). Using the rules that rays passing through the optical centre of a lens pass undeviated, while those parallel to the optic axis of the lens must pass through the conjugate focal point of the lens, the ray diagram for the microscope can be drawn (Fig. 3). From the thin lens equation, $\frac{1}{f} = \frac{1}{u} + \frac{1}{v}$; where f is the focal length, u = the object distance, and v = the image distance; and using the real-is-positive convention, we find that real magnified images are only produced when the object is between one and two focal lengths, f_o , in front of the objective.

The objective produces a real inverted magnified image of the object AB at A' B', which acts as the object for the eyepiece. It is arranged that A' B' lies just within the object focal distance, f_e , of the eyepiece. Consequently, when placed at the exit pupil of the eyepiece, the eye sees a virtual image at A'' B''. The virtual image A'' B'' is magnified but not reinverted and its exact position depends on how the microscope is focused. Usually we assume that it is focused within the image at the nearest distance of distinct vision, which is about 250 mm.

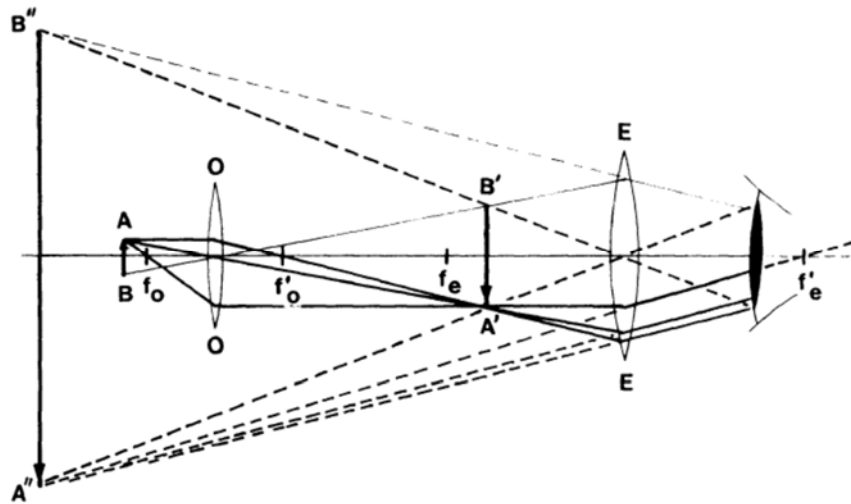


Figure 3: Ray diagram illustrating image formation in the compound microscope.

The magnification M produced by a lens is given by the equation,

$$M = \frac{v}{u}$$

Substituting u from the thin lens equation, for real image,

$$M = \frac{v}{f} - 1$$

Therefore, for objective lens (that produces real image),

$$M_o = \left(\frac{v_o}{f_o} - 1 \right) = \left(\frac{v_o - f_o}{f_o} \right)$$

here the subscripts O refer to the objective, and since the optical tube length (Fig. 1a), $t = v_o - f_o$,

$$M_o = \left(\frac{t}{f_o} \right)$$

Hence, the magnification produced by the objective is fixed and depends on the focal length of the objective lens. For eye piece, that forms a virtual image,

$$M_e = \left(\frac{v_e}{f_e} + 1 \right) = \left(\frac{v_e + f_e}{f_e} \right)$$

where the subscripts e refer to the eyepiece. Assuming that the image is formed at the nearest distance of distinct vision, D, we can put, $D = v_e + f_e$,

$$M_e = \left(\frac{D}{f_e} \right)$$

As in the case of the objective, the magnification produced by the eyepiece is fixed and depends on its focal length. Hence the total magnification, M_{total} , produced by the microscope is

$$M_{total} = M_o \times M_e = \left(\frac{tD}{f_o f_e} \right)$$

However, if a tube lens is incorporated in the microscope the total magnification is

$$M_{total} = M_o \times M_e \times M_t$$

where M_t is the magnification produced by the tube lens.

3. Mechanisms of image formation

Geometrical optics enables us to understand the way in which a microscope produces a magnified image, but it tells us nothing about the quality of the image that is formed. To understand this, we must consider image formation in terms of the wave theory of light.

We can consider any point on an object to be a light source emitting waves in all directions, and an object to be made of many such sources. If all the waves leaving a point on the object could recombine, a perfect image of the point would form. In practice this cannot be achieved because many of the waves travel in directions from which they cannot be collected by the lens forming the image and are lost (Fig. 4). Thus the perfection of the image depends on the amount of light from the object collected by the optical system, provided that the rays are focused accurately and are in phase. The phase relationship of the rays is especially important when they are coherent. When two coherent rays¹ combine, the amplitudes of their waves are additive. Hence, since

¹ Waves having same frequency are able to interfere and are described as being coherent, while those that cannot are described as being non-coherent. Truly coherent light, in which the waves are all exactly in phase.

the brightness of the light is proportional to the square of the amplitude of the waves, when two waves of equal amplitude are exactly in phase the brightness of the resultant wave is four times that of either of the original waves. Conversely, when they are exactly out of phase the light is extinguished. On the other hand, non-coherent waves do not interfere or combine. Thus when two non-coherent waves of equal amplitude propagate, the intensity is simply twice that of either of the original waves. When, in an image formed by coherent light, the waves fail to remain in phase the quality of the image deteriorates, and the image may be destroyed. Waves remain in phase when their optical paths contain the same number of waves. For example, consider the light passing through a convex lens; the axial ray has the shortest path through air and the longest path through glass, while the marginal rays have the longest path through air and the shortest path through glass. The wavelength in glass is shorter than that in air, thus in principle the lens is designed so that all paths are optically equal. Nevertheless, in practice it is sufficient that the converging rays remain in phase within a quarter of a wavelength.

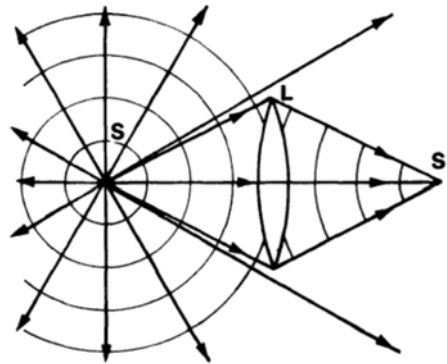


Figure 4: Sketch showing why a lens is unable to form a perfect image.

Diffraction is the most important image-forming mechanism. It occurs when light passes an edge and causes the shadow to have a diffuse edge. Consider a wide slit illuminated with coherent (monochromatic) light (Fig. 5). As the light emerges from the slit it spreads sidewise. When the edges of the slit are brought closer together the spread becomes greater and eventually light and dark side bands appear on either side of the direct beam. In a similar manner, when a pinhole is illuminated the direct beam is surrounded by concentric alternate dark and light rings, known as the Airy disc. The direct beam is called the zero-order beam, and successive diffracted beams are described as first-order, second-order and so on. The effect of a single slit can be reinforced by those of other regularly spaced slits with an interslit spacing equal to the width of the slit.

2.2.1 Abbe's theory of the microscope

Consider the case of image formation in the transmitted-light microscope. In Fig. 6 a transparent object A with a fine two-dimensional periodic structure is illuminated by a collimated (parallel) beam of light. The structure diffracts light, as described previously. Thus, from a point P₀ the objective, O, collects the direct beam and some of the diffracted beams. Since the object is illuminated with collimated light the direct and diffracted beams are also collimated. Consequently, after passing through the objective all the beams are brought to focus in the back focal plane of the objective, F, each beam forming an image of the light source, S. These images of the light source are Huyghenian subsidiary sources, each of which sends out waves that form by destructive interference the image of the point P₀ in the image plane I at point P_i. The subsidiary sources lie on the arc of a circle centred at P_i, so that all the subsidiary waves reach P_i in phase and reinforce each other. Thus, there is produced a bright image of P₀ that is surrounded by alternate dark and light rings formed by

interference. The central bright disc is the true image of P0 but the unavoidable surrounding bright rings reduce its sharpness.

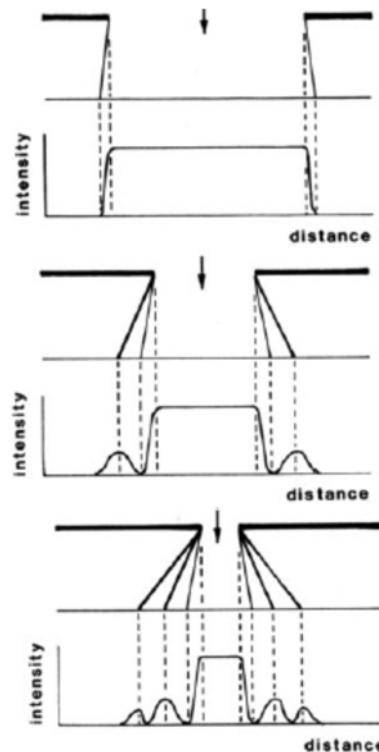


Figure 5: Diffraction of light at a slit.

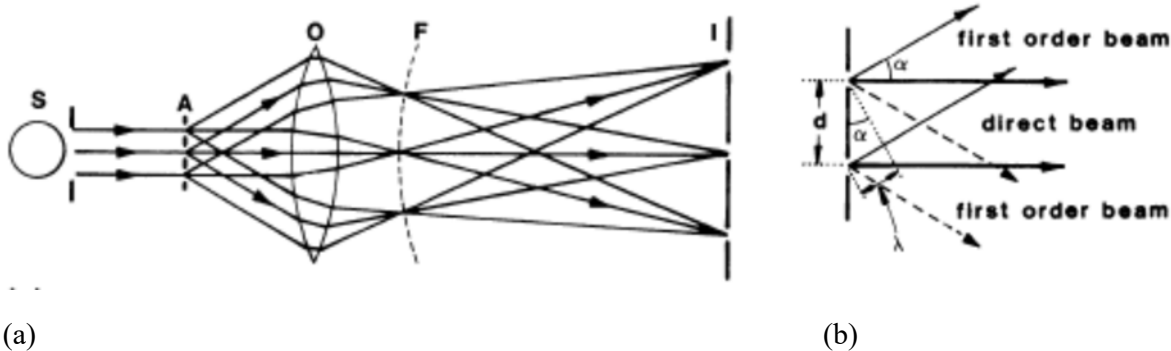


Figure 6: (a) Abbe's theory of image formation in the microscope. (b) Criterion for the formation of first-order diffracted beams at the object

We may draw three main conclusions from the detailed theory. Firstly, at least two beams must be collected by the objective if the detail in the image is to be resolved, because at least two beams are needed for interference to occur. (Fig. 7a) If only the direct beam is collected no detail is shown in the image (Fig. 7b). Secondly, to form the best possible image all the beams diffracted by the object ought to be collected by the objective and used to form the image. Thirdly, the sharpness and intensity of the central bright disc depend upon collecting the direct beam and an optimum number of diffracted beams. (Fig. 7c) In discussing the fringes produced by parallel slits we noted that the angular displacement of successive spectra was greater the smaller the spacing of the slits. Similarly, we find that the finer the diffracting structure the greater become the angles between the direct beam and the diffracted beams. Consequently, the finer the diffracting structure the wider

must be the angle of the cone of acceptance of the objective if the first-order diffracted beams are to be collected. Moreover, the greater the angle of the cone of acceptance, which is measured by the numerical aperture, the more likely are higher-order diffracted beams to be collected by the objective, thus forming a more accurate image. (Fig. 7c)

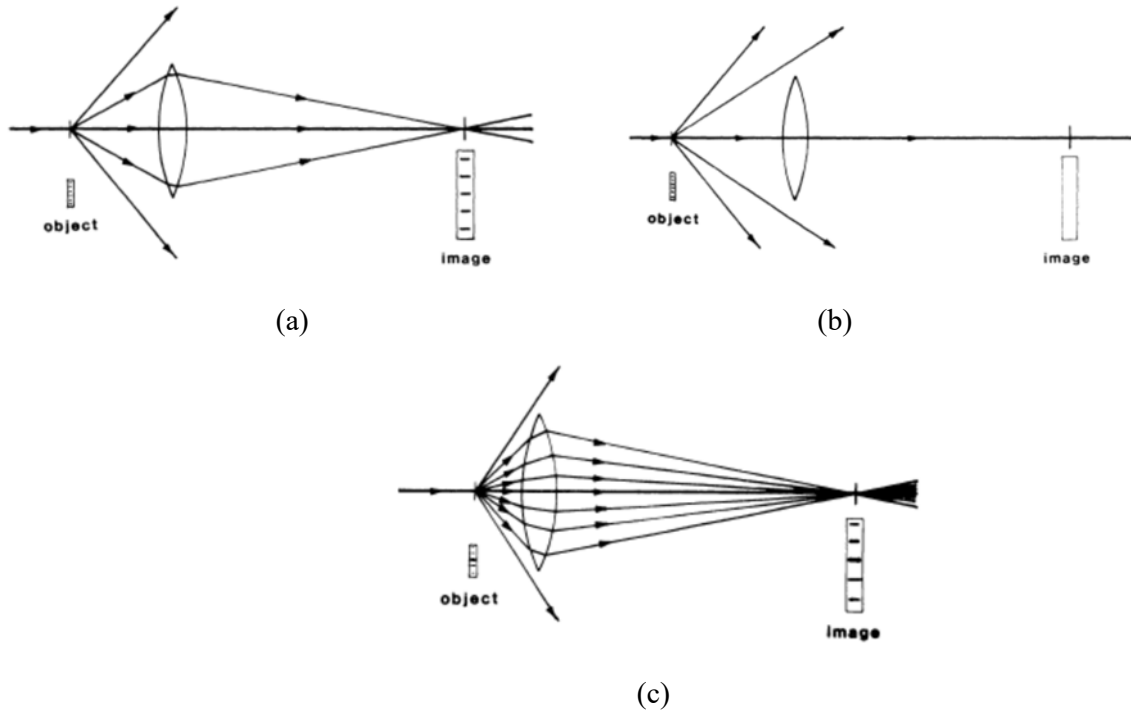


Figure 7: (a) Direct and first-order diffracted beams are collected by the objective. Detail in the image is resolved. (b) Only the direct beam is collected by the objective. Detail in the image is not resolved. (c) The direct and several orders of diffracted beams are collected by the objective. Detail is resolved accurately.

If the angle between the direct beam and the first-order diffracted beams is so big that the latter cannot be collected by the objective when the objective is illuminated normally (Fig. 7b), it may be possible to enable the objective to collect the direct beam and a diffracted beam by illuminating the object obliquely (Fig. 8a)

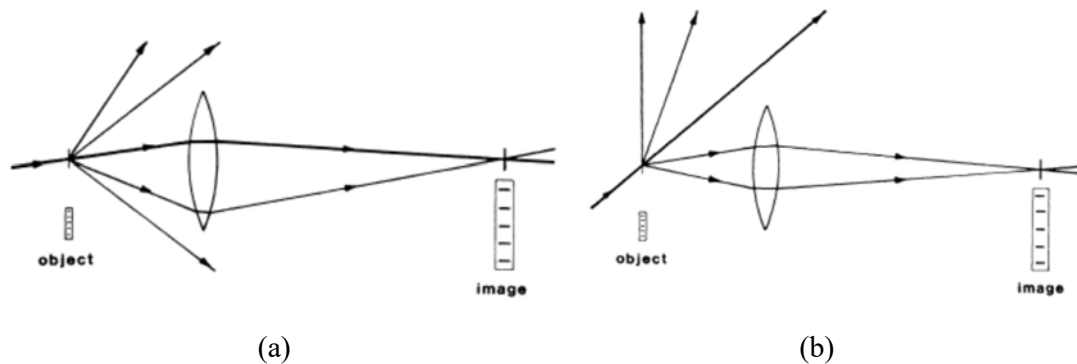


Figure 8: (a) By illuminating the object in Fig. 7b obliquely the objective collects the direct beam and a diffracted beam. Resolution of detail in the image is restored. (b) The direct beam is so oblique that it no longer enters the objective, but two diffracted beams are collected by the objective. Detail in the image is resolved, but with reversed contrast.

This permits an image to be formed but its quality is poor. As the obliquity of the illuminating beam increases, the ability of the objective to resolve detail increases because the angle between the direct beam and the first-order diffracted beam that the objective can accept increases. Eventually we reach a point at which the direct beam no longer enters the objective, so that only the diffracted beams are collected. However, provided that two or more diffracted beams are collected an image is formed (Fig. 8b). This technique produces reversed contrast and is described as dark-ground illumination.

While diffraction is the essential mechanism of image formation in the microscope certain other mechanisms may also be involved. These are reflection, absorption, refraction, fluorescence, and polarization. Reflection is an essential mechanism in the formation of images of metallic and other opaque specimens. It occurs at interfaces at which there is a large difference in the refractive indices of the media, as between air and metal. In the metallurgical microscope the specimen may be illuminated either vertically, i.e. normal to the surface of the specimen along the optic axis of the microscope, or very obliquely, producing bright-field and dark-field illumination respectively. In the former, light from the surfaces of grains that are perpendicular to the incident beam is reflected back into the objective, while light from features in the microstructure, such as grain boundaries and tilted areas of the surface, is reflected away from the objective, so that they appear dark. Conversely with dark-field illumination, light reflected from surfaces which are almost perpendicular to the optic axis of the microscope is reflected away from the objective, while light from features in the microstructure is reflected into it.

4. Magnification, resolution and perception

The image formed by a microscope is produced by the action of both the objective and the eyepiece. Of the two lenses, the objective is most important because it determines the detail that can be seen in the final image and the optimum useful magnification.

4.1 Objective magnification

The magnification produced by an objective is determined by its focal length and the optical tube length. For a given tube length the focal length determines the object distance and the primary magnification M , which is given by the simple equation,

$$M_o = \left(\frac{t}{f_o} \right)$$

where t = optical tube length, and f_o = focal length of the objective.

4.2 Limit of resolution and resolving power

The limit of resolution is a measure of the ability of the objective to separate in the image individual, adjacent details that are present in the object; it is the distance between two points in the object that are just resolved. The resolving power is the reciprocal of the limit of resolution and is expressed in units of lines resolved per millimetre.

In section 3 we saw that for resolution to occur at least both zero- and first-order diffracted beams must be collected by the objective. Suppose a grating with rulings of spacing d is illuminated in vacuum at an angle of incidence i by an oblique beam of coherent, monochromatic light of wavelength λ_0 , and that the zero- and first-order beams are just collected by the objective. (Fig. 9a) Since the path difference between zero- and first-order diffracted rays from successive rulings in the grating is exactly one wavelength (Fig. 9b)

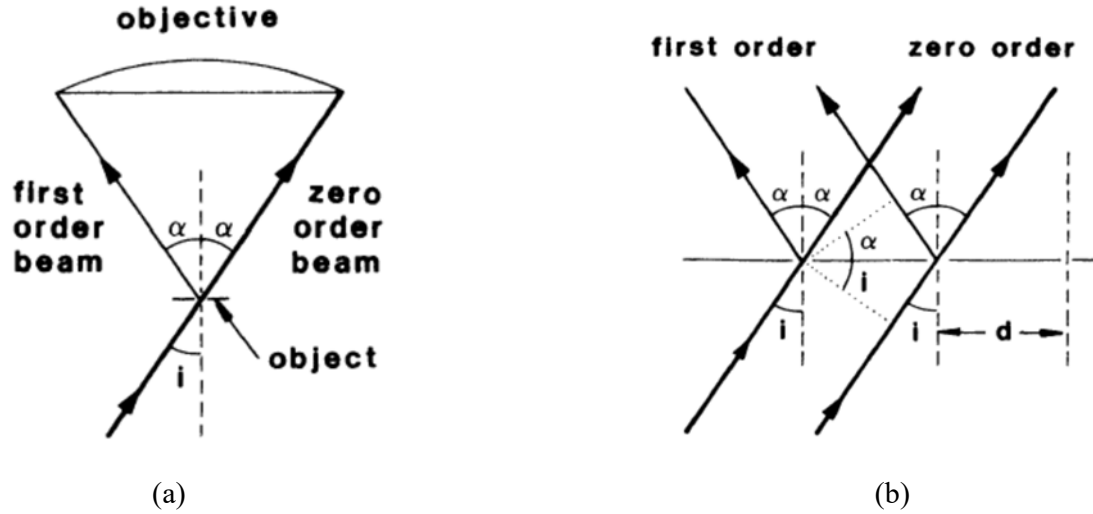


Fig. 9 (a) Collection of zero- and first-order diffracted beams by the objective (b) The condition for the diffraction of first-order beams by a ruled grating.

$$d \sin i + d \sin \alpha = \lambda$$

where 2α is the angle through which the first-order beam is diffracted. However, since the two beams are just collected by the objective i is equal to α , thus the limit of resolution, d_{min} is

$$d_{min} = \frac{\lambda}{2 \sin \alpha}$$

Moreover, if the object space is filled with a medium of refractive index n , the wavelength of the light in the medium, λ_n is

$$\lambda_n = \frac{\lambda}{n}$$

Therefore, when an objective is used with the object space filled with a medium of refractive index n ,

$$d_{min} = \frac{\lambda_n}{2 \sin \alpha} = \frac{\lambda}{2n \sin \alpha} = \frac{\lambda}{2(NA)}$$

$n \sin \alpha$ is called the numerical aperture, NA, of the objective, and is one of its important characteristics.

Rayleigh arrived at the criterion for resolution using a complementary approach. Owing to diffraction, the image of a point of light is a point surrounded by concentric dark and light rings, the intensities of which decrease rapidly. When the images of two points are brought sufficiently close together for their diffraction patterns to overlap, the points cease to be distinguishable. Rayleigh assumed that this occurs when the separation of the points in the image is equal to the radius of the first dark ring, i.e. when the first dark ring of point 1 coincides with the central bright disc of point 2, and vice versa (Fig. 10). Quantitatively, this leads to an expression for the limit of resolution,

$$d_{min} = \frac{k\lambda}{2(NA)} = \frac{0.61 \lambda}{(NA)}$$

where k has a value of 1.22 when the objective is filled with light from a suitable condenser. The value is close to that deduced from diffraction theory.



Figure 10: Overlapping intensity distributions for the images of two points.

From the above, we see that the resolution can be increased either by increasing the NA or by decreasing the wavelength of the light used to illuminate the specimen. Consequently, microscopes usually are provided with a series of objectives with NA values in the range 0.1 to 1.3; the latter value is close to the practical upper limit for NA. The effect on resolution of increasing the NA from 0.25 to 0.55 is shown in Fig. 11.

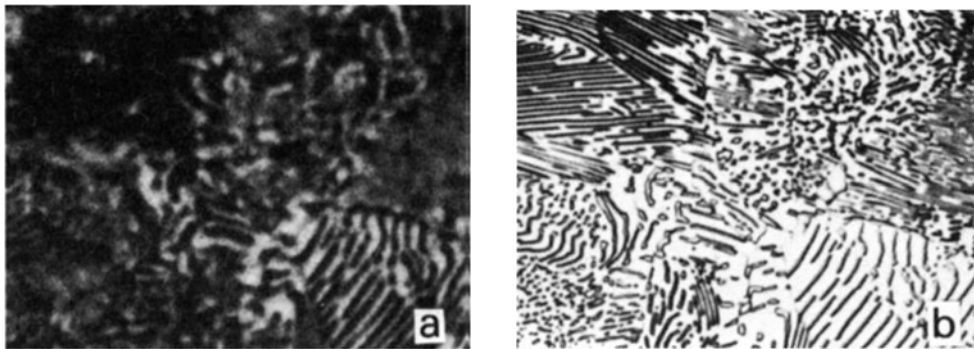


Figure 11: Effect of NA on the resolution of pearlite in green light, bright-field illumination. (a) NA = 0.25, magnification x 500 (2000 x NA), resolution has failed except at the lower left hand corner; (b) NA = 0.55, magnification x 500 (900 x NA), same field as (a). the lamellar structure is largely resolved.

Visible light ranges in wavelength from ~ 650 nm at the red end of the spectrum to ~ 400 nm at the blue end. Therefore, the limit of resolution at the blue end is about one and a half times better than at the red end. However, the eye is very insensitive to blue light, so that the improvement in resolution is obtained at the expense of poor visibility. The eye is most sensitive to light in the yellow-green part of the spectrum, i.e. of wavelengths 520- 600 nm, and in practice light which is rich in these wavelengths, e.g. daylight or light from a tungsten filament lamp, is often used for visual examination. Further improvement in resolution by up to a factor of two can be achieved by the use of ultraviolet light of shorter wavelengths than visible light. However, focusing ultraviolet light is difficult and optical glasses strongly absorb wavelengths less than ~ 300 nm. Consequently, either objective lenses and eyepieces made from quartz glass, which transmits wavelengths down to ~ 200 nm, or catoptric (mirror) objectives, which also overcome the focusing difficulty, must be used.

4.3 Perception

Resolution is concerned with the ability of the objective to separate in the image features that are close together. However, features can be perceived when they cannot be resolved. Thus the image of a single pinhole

broadens when the NA is small or the wavelength of the light long, because more concentric dark and light rings are formed around the bright disc, but the image is not destroyed. Moreover, when the NA is increased more light is diffracted into the central disc, the contrast is increased and the image is perceived more easily. The theory of perception is complex, but it shows that the limit of perception usually is between a tenth and a hundredth of the limit of resolution. In practice, it is useful to know that a feature is present, even though its detail cannot be resolved.

4.4 Depth of field and field of view

When a lens is sharply focused on a given plane, points which lie a little further from or closer to the objective also appear in acceptable focus. The range of distances over which points appear in acceptable focus is called the depth of field. Thus the object plane is really a thin layer in which all points appear to be in focus. Qualitatively, this occurs because points in the image are Airy discs, not perfect discs of light. Consider two parallel planes P 1 and P 2 which are close together and perpendicular to the optic axis of the objective (Fig. 12), and suppose that the objective is sharply focused on plane P 1, producing an image at I. Rayleigh suggested that for plane P 2 also to appear just in focus the paths in the image space of the marginal rays from the two planes should differ by less than a quarter wavelength. If the difference is greater the bundle of rays from a point P 2 produces an image point that is greater in diameter than the circle of confusion arising from diffraction.

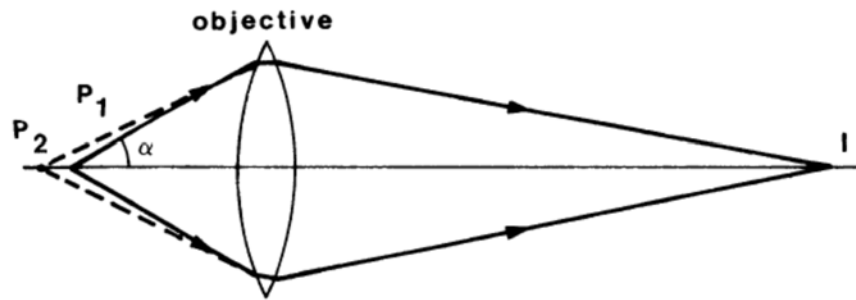


Figure 12: The principle of depth of field of an objective

Using this criterion the depth of field, h , can be shown to be,

$$h = \pm \frac{\lambda/4}{n \sin^2 \alpha} = \pm \frac{n\lambda}{(NA)^2}$$

where $\lambda/4$ = the maximum allowable path difference which determines the depth of field, and n = the refractive index of the object space. In practice, the equation gives a conservative estimate of the depth of field. The calculated effect of NA on the depth of field for dry and immersion objectives is shown in Table 1.

Table 1: Effect of numerical aperture on depth of field

Numerical aperture		0.10	0.25	0.50	0.75	1.00	1.25
Depth of field (μm).	Air, $n = 1.0$	56	8.6	2.0	0.81	(0.27)	—
	Oil, $n = 1.5$	82	13	3.2	1.4	0.73	0.42

Wavelength of light 550 nm.

The depth of field is limited, especially with objectives of high NA. This is important for three practical reasons. Firstly, focusing becomes more difficult the smaller the depth of field. Accordingly, with a series of parfocal objectives it is good practice to focus first with the objective of lowest NA, because there is less chance of missing the plane of focus. Secondly, it is difficult to examine objects with irregular surfaces, e.g. fracture surfaces, at higher magnifications because it is impossible to obtain all the surface in focus simultaneously. Thirdly, mounting the surface to be examined perpendicular to the optic axis of the microscope becomes more critical the smaller the depth of field.

4.5 Optimum useful magnification

To make proper use of an objective we need to know how much the image must be magnified so that we can perceive all the detail in it. This depends on the ability of the eye to resolve detail. The unaided eye is able to just distinguish between two points if they subtend an angle of one minute of arc at the eye, which corresponds to a separation of $\sim 100\ \mu\text{m}$ at the nearest distance of distinct vision. However, for working without eye strain the separation needs to be two to four times larger. The minimum magnification needed for the eye to just distinguish between adjacent points is the ratio of the minimum separation of two points that can be just distinguished by the eye to the resolution of the objective. Thus, we compare a low-power dry objective and a high-power oil immersion objective in Table 2, assuming the use of yellow light. Comparison of the data for the two objectives shows that the ratio of optimum magnification to objective NA is about 1000, and a useful rule of thumb is that the overall magnification should be 500 to 1000 times the objective NA. Moreover, both require the use of eyepieces of the same magnifying power to achieve the optimum magnification; this is generally true for all objectives. While it is necessary for the overall magnification to exceed a minimum value if we are to see all the resolved detail, excessive magnification should be avoided. It increases the size of the image without increasing the amount of detail and may degrade its quality; it is called empty magnification

4.6 Defects in lenses

The quality of the final image is no better than that of the primary image produced by the objective. In practice, there are several defects that can mar the quality of the image produced by a lens, especially an objective, the most important of which are spherical aberration, chromatic aberration, astigmatism, coma, curvature of field, and distortion.

Table 2 Estimation of the overall magnification required to enable resolved detail to be seen

	Low-power dry objective	High-power oil-immersion objective
Wavelength of light	580 nm	
Numerical aperture	0.15	1.30
Focal length	16 mm	1.8 mm
Objective magnification	$\times 10$	$\times 95$
Resolution ($\lambda/2 \text{ NA}$)	1935 nm	223 nm
Minimum magnification	$10^{-4}/1.935 \times 10^{-6}$ $= 52.5$	$10^{-4}/2.23 \times 10^{-6}$ $= 460$
Minimum eyepiece magnification	$52.5/10 \simeq 5$	$460/95 \simeq 5$
Optimum magnification ($\times 2$ to $\times 4$ minimum magnification)	100 to 200	1000 to 2000
Optimum eyepiece magnification	10 to 20	10 to 20
Optimum magnification/NA	660 to 1300	770 to 1500

4.6.1 Spherical aberration

When monochromatic light rays from a point source pass through the outer parts of a lens they are brought to a focus at a shorter distance from the lens than are those that pass through the central parts (Fig. 13), owing to the more oblique rays being refracted through a greater angle than those near the optic axis. The aberration is caused by the spherical shape of the lens surfaces, hence spherical aberration, and it is more severe the greater the aperture of the lens.

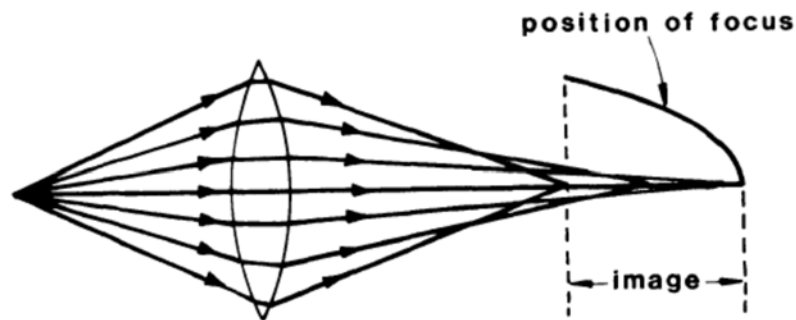


Figure 13: Spherical aberration. Rays passing through different lens zones do not focus at a point. The curve indicates the amount of spherical aberration

It occurs for most positions of an axial object point, but for certain positions it becomes zero. Such aberration-free object and image points are called aplanatic points. This aberration can be largely, but not completely, eliminated by use of combinations of converging and diverging lenses of different refractive index. These corrections are also effective for coma, which is a form of asymmetrical spherical aberration.

4.6.2 Chromatic Aberration

When white light is focused by a lens, light of different wavelengths is brought to focus at different distances from the centre of the lens (Fig. 14), violet light being focused closer to the lens than red light. Consequently a simple lens produces several images of the object in different colours, each focused in a slightly different plane, causing the image to have coloured fringes around its outline.

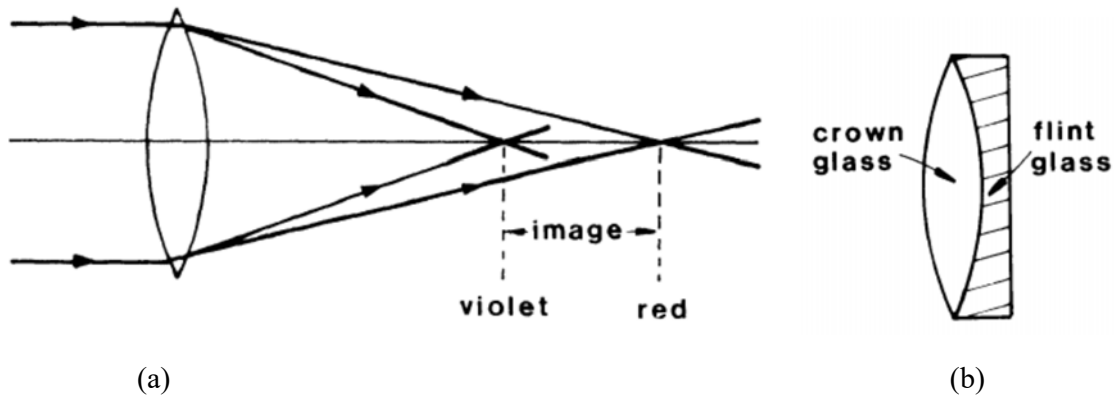


Figure 14: (a) Longitudinal chromatic aberration. Light of shorter wavelengths is brought to focus closer to the lens than light of longer wavelengths, causing the image to be surrounded by colour fringes. (b) An achromatic doublet

This occurs because the refractive index of a transparent isotropic material is greater for light of shorter wavelength than for light of longer wavelength. The proportion in which the refractive indices differ is called the dispersion of the material. Strictly, it is measured by the slope of the curve of refractive index versus wavelength and is greater for shorter wavelengths than for longer wavelengths. Different kinds of glass have different reciprocal relative dispersions, e.g. crown glass ~ 60 and flint glass ~ 38 . Consequently a converging lens of crown glass can be combined with a weaker diverging lens of flint glass (Fig. 3.15), so that the chromatic aberrations cancel for certain wavelengths, while leaving some converging power: the combination is called an achromatic doublet.

4.6.3 Astigmatism and curvature of field

Astigmatism is a defect in which the images of points off the optic axis are drawn out into blurred lines or discs. It increases with distance from the optic axis and causes poor definition of images formed away from the axis. The extent of the aberration can be reduced by stopping down the lens, but this impairs resolution. The cause of astigmatism is illustrated in Fig. 15; the tangential plane is defined by the vertical diameter, TT' , of the lens, while the sagittal plane is defined by the horizontal diameter, SS' . The tangential rays OT and OT' produce the tangential image, I_t , and the sagittal rays OS and OS' produce the sagittal image, I_s , that lie on surfaces of different curvature. The horizontal line image, I_t , lies in the sagittal plane and the vertical image I_s , in the tangential plane. The circle of least confusion lies between the two images and is the best image possible. Correction for astigmatism causes the two images to coincide, but they continue to lie on a curved surface.

Curvature of the image field arises from the change in focal length of the lens as the position of the point on the lens moves away from the optic axis; it depends on the lens geometry and refractive index. The radius of curvature of the field is approximately equal to the focal length of the objective. Consequently it is more apparent in images formed by high-power objectives. It cannot be eliminated in objectives of conventional design, but is largely eliminated in modern flat-field objectives.

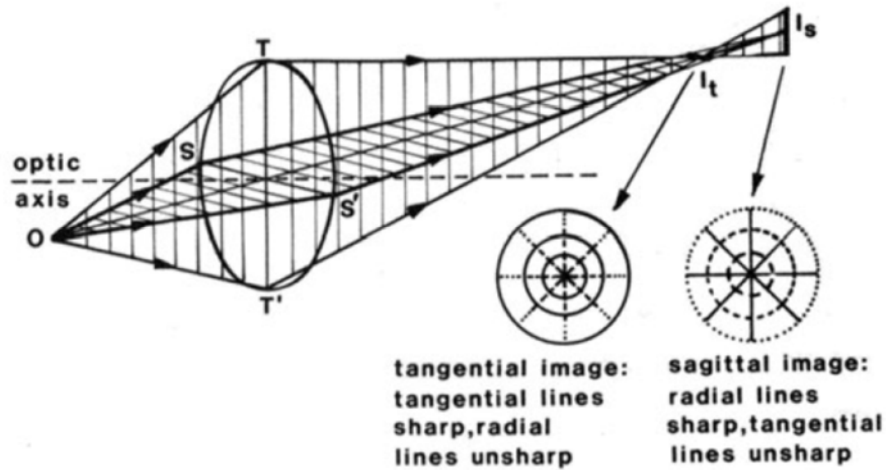


Figure 15: Astigmatism. Rays lying in the tangential plane are brought to a different focus from those lying in the sagittal plane.

4.6.4 Coma and distortion

Coma causes the image of a non-axial point to be reproduced as an elongated comet shape, lying in a direction perpendicular to the optic axis. It is a form of asymmetrical spherical aberration affecting non-axial object points. The comet is made up of a series of overlapping circular images (Fig. 16) that increase in size in going from I₁ and I₃. It results from differences in magnification arising from rays meeting the lens at widely differing angles. The overlapping images decrease in intensity from I₁ and I₃. As with astigmatism, it can be reduced by stopping down the objective.

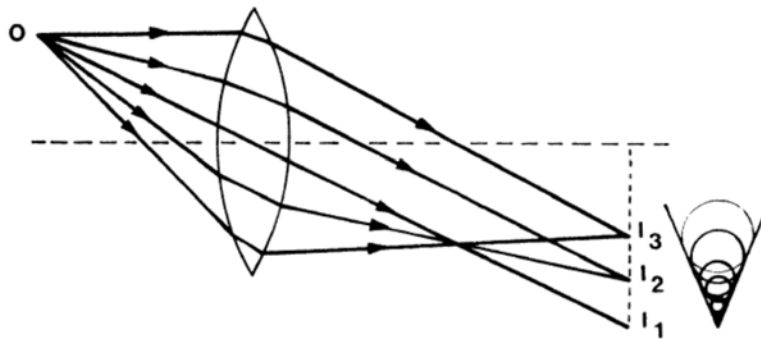


Figure 16: Coma. Oblique incident rays which pass through the outer parts of the lens do not converge to the same point as those passing through the central parts.

Distortion arises from variation in magnification with distance of the object point from the optic axis. It causes either barrelling or pincushioning (Fig. 17), depending on whether the magnification decreases or increases with distance from the optic axis. It occurs in both objectives and eyepieces, but is more common in the latter, and is difficult to eliminate completely.

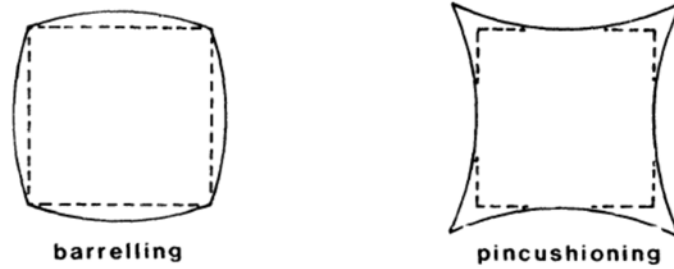


Figure 17: Distortion. (a) Barrelling and (b) pincushioning

Reference

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Exercise

1. In an optical microscope, the optical tube length is 150 mm, and distance of distinct vision is 100 mm, focal length of objective lens is 150 mm, and focal length of eye piece is 10 mm, calculate the total magnification of the microscope.
2. The refractive index of the objective lens is 50.5, and refractive index of condenser lens is 24.9, and semi apex angle is 10 degrees, the wavelength of the light used is 250 nm. If all these parameters belong to a microscope, will they give optimum performance in the bright field imaging? also calculate the resolving power of the microscope.
3. Find out the relationship between the resolution limit ($d_{\min}$) of green and blue light in the visible spectrum. Consider, refractive index (n) of the medium for green and blue light are 1.51 and 1.54, respectively and the semi-apex angle is 10° .
4. Estimate the depth of focus (h) for the given parameters of the objective lens of an optical microscope; refractive index 1.7, wavelength of light, $\lambda = 550$ nm, semi-aperture angle, 45° . Establish the effect of numerical aperture on depth of focus (h) and the limit of resolution.